arXiv:2605.18910v1 [stat.ME] 17 May 2026

A Tutorial on Symbolic Structural Identifiability Analysis of ODE Models in Julia

Abdallah Alsammani*
Department of Mathematical Sciences, Delaware State University, Dover, DE 19901

Abstract

Structural identifiability analysis determines whether the parameters of a mechanistic ordinary differential equation (ODE) model can be uniquely recovered from ideal observations and is therefore a fundamental prerequisite for reliable parameter estimation. This tutorial presents a modern, reproducible computational framework for symbolic structural identifiability analysis using the Julia package `StructuralIdentifiability.jl`. We provide a rigorous yet accessible introduction to local and global identifiability, observability, parameter-to-output mappings, and identifiable parameter combinations, together with a unified workflow based on the core functions `@ODEmodel`, `assess_local_identifiability`, `assess_identifiability`, and `find_identifiable_functions`. The framework is demonstrated through seven case studies from epidemiology, pharmacokinetics, and systems biology, illustrating globally identifiable systems, local-only identifiability, structural non-identifiability, and recovery of identifiability through additional measurements and reparameterization. Beyond the theoretical foundations, the tutorial emphasizes practical model reformulation, experimental design, and reproducible scientific workflows within the Julia SciML ecosystem, providing a comprehensive reference for researchers and graduate students working with mechanistic ODE models.

Keywords: structural identifiability; ordinary differential equation models; symbolic computation; Julia; `StructuralIdentifiability.jl`; reproducible scientific computing.

1. Introduction

1.1. Structural identifiability and mechanistic modelling

Mechanistic ordinary differential equation (ODE) models constitute one of the principal mathematical frameworks for describing dynamical processes in epidemiology, systems biology, pharmacokinetics, ecology, neuroscience, and biomedical engineering. Such models encode scientific hypotheses regarding the mechanisms governing the temporal evolution of biological, chemical, or physical systems through interacting state variables and parameters. Once calibrated to experimental or observational data, mechanistic models are routinely employed for forecasting, intervention analysis, uncertainty quantification, optimal experimental design, and hypothesis testing [8, 10, 1].

The reliability of every downstream inference derived from a calibrated model depends fundamentally on whether the unknown model parameters can, in principle, be uniquely recovered from the available observations. This question is addressed by *structural identifiability theory*, which studies the uniqueness of parameter recovery under the assumption of ideal noise-free continuous measurements.

*Email: `aalsammani@desu.edu`.

A parameter is said to be structurally identifiable if distinct parameter values cannot generate identical model outputs [5, 9]. When structural identifiability fails, parameter estimation procedures may produce multiple indistinguishable solutions that fit the data equally well, leading to non-unique parameter estimates, inflated uncertainty, ill-conditioned confidence intervals, and potentially misleading biological interpretation [1, 18, 27].

Structural identifiability should be distinguished from *practical identifiability*. Structural identifiability is an intrinsic property of the model equations and observation scheme, whereas practical identifiability concerns the numerical recoverability of parameters from finite, noisy, and discretely sampled data. Consequently, a model may be structurally identifiable yet practically unidentifiable because of insufficient data quality, poor experimental design, parameter correlations, or limited observation windows [1, 28, 29]. Structural identifiability is therefore a necessary prerequisite for meaningful parameter estimation, since no amount of additional data can recover parameters that are structurally non-identifiable under a fixed measurement scheme.

The identifiability properties of an ODE model depend not only on the model equations themselves, but also on the available measurements, known inputs, and assumptions regarding initial conditions. In many applications, especially in epidemiology and systems biology, only a subset of the state variables can be observed directly. Latent compartments, partially observed populations, unknown initial conditions, and aggregated measurements frequently introduce hidden parameter symmetries and non-identifiable parameter combinations. Consequently, structural identifiability analysis plays a central role in model formulation, measurement selection, and experimental design [10, 4, 12].

1.2. Challenges in epidemiology, systems biology, and biomedical modelling

Identifiability issues are particularly pronounced in biomedical and epidemiological modelling. Compartmental epidemic models commonly include latent populations such as exposed, asymptomatic, hospitalized, or environmental compartments that cannot be observed directly. Similarly, systems biology and pharmacokinetic models often involve hidden molecular species, indirect measurements, nonlinear interaction terms, and unknown initial conditions. In many practical settings, available data correspond only to incidence-like aggregated quantities, rather than direct measurements of the full system state.

These limitations can lead to severe parameter non-identifiability, even in comparatively simple models. Classical examples include SEIR-type epidemic systems, within-host viral dynamics models, vector-host transmission systems, and multi-compartment pharmacokinetic models, where multiple parameter combinations may generate identical observable trajectories [1, 4, 2]. As a result, parameter estimates may depend strongly on optimization algorithms, prior assumptions, regularization choices, or initial guesses rather than on the information content of the data itself.

The increasing adoption of scientific machine learning, neural ordinary differential equations, differentiable programming, and hybrid mechanistic–data-driven modelling frameworks has further amplified the importance of rigorous identifiability analysis. In these settings, structurally non-identifiable parameterizations may produce unstable latent representations, ambiguous mechanistic interpretation, and misleading uncertainty estimates despite excellent predictive performance [30, 20]. Consequently, modern computational modelling workflows increasingly require identifiability analysis as an essential preliminary step before parameter estimation, Bayesian inference, or uncertainty quantification.

Structural identifiability analysis provides a mathematically rigorous framework for diagnosing

these issues before model calibration. Importantly, identifiability analysis is not merely diagnostic, but also constructive. In addition to determining which parameters are uniquely recoverable, it identifies combinations of parameters that remain observable despite individual parameter non-identifiability. These identifiable combinations frequently suggest natural reparameterizations, additional measurements, or experimental redesigns that restore identifiability and improve model interpretability [16, 29, 19].

1.3. Symbolic identifiability methods and computational approaches

The mathematical theory of structural identifiability has developed over more than five decades, beginning with the foundational work of Bellman and Åström [5], Pohjanpalo [6], Cobelli and DiStefano [7], Walter and Pronzato [8], Ljung and Glad [9], and Miao and colleagues [10]. A broad range of analytical and computational methodologies has since been developed for assessing parameter identifiability in nonlinear dynamical systems.

Classical approaches include Taylor series expansion methods [6], similarity transformation techniques for linear systems, differential algebra approaches based on input-to-output elimination [9, 11], observability-based Lie derivative methods [12], and probabilistic algebraic approaches relying on Gröbner bases or differential elimination [13, 14]. These methodologies have led to the development of several influential software packages for symbolic identifiability analysis, including **DAISY** [11], **GenSSI** [15], **STRIKE-GOLDD** [12], **COMBOS** [16], and **SIAN** [17, 13].

Despite major algorithmic advances, symbolic structural identifiability analysis remains underutilized in applied scientific modelling. Several factors contribute to this limitation. First, many existing software platforms depend on proprietary computer algebra systems such as Maple or MATLAB, which restrict reproducibility and complicate integration with modern open-source computational workflows. Second, many symbolic methods require specialized mathematical expertise, creating a substantial barrier for researchers whose primary focus lies in biological or biomedical applications. Third, large-scale nonlinear models often present considerable computational challenges, especially for global identifiability analysis and the computation of identifiable parameter combinations [14, 13].

Recent developments within the Julia scientific machine learning (SciML) ecosystem have substantially improved the accessibility and reproducibility of symbolic identifiability analysis. `StructuralIdentifiability.jl` [14, 19] is a modern open-source Julia package that integrates symbolic identifiability analysis directly into contemporary computational modelling workflows. The package implements state-of-the-art algorithms for local identifiability using the Sedoglavic probabilistic method [22], global identifiability analysis via SIAN-style differential algebraic approaches [17, 13], and the computation of identifiable parameter combinations through differential elimination techniques [14, 19].

Unlike many earlier identifiability platforms, `StructuralIdentifiability.jl` interoperates naturally with the broader Julia SciML ecosystem, including `ModelingToolkit.jl`, `DifferentialEquations.jl`, `Catalyst.jl`, and related packages [20, 21]. This interoperability enables symbolic analysis, numerical simulation, automatic differentiation, parameter estimation, uncertainty quantification, and reproducible scientific computing within a unified open-source environment.

1.4. Objectives and contributions of this tutorial

This article is intended as a comprehensive computational tutorial and reproducible reference for symbolic structural identifiability analysis within modern ODE modelling workflows. The tutorial is

designed for researchers, graduate students, and applied scientists working in mathematical biology, epidemiology, pharmacokinetics, systems biology, and scientific machine learning who seek a rigorous yet practical framework for identifiability analysis of mechanistic dynamical systems.

The primary objective of this tutorial is to bridge the gap between the mathematical theory of structural identifiability and its practical implementation within reproducible computational workflows. Rather than focusing on new theoretical identifiability results, the article emphasizes computational methodology, interpretation, and practical application. Specifically, this tutorial:

1. provides a rigorous yet accessible introduction to structural identifiability, observability, parameter-to-output mappings, and identifiable parameter combinations for nonlinear ODE models;

2. presents a unified computational workflow for symbolic identifiability analysis in `StructuralIdentifiability.jl`, centered on the functions `@ODEmodel`, `assess_local_identifiability`, `assess_identifiability`, and `find_identifiable_functions`;

3. demonstrates the methodology through a progression of case studies drawn from epidemiology, pharmacokinetics, viral dynamics, and systems biology, illustrating globally identifiable systems, locally identifiable systems, structural non-identifiability, identifiable parameter combinations, and recovery of identifiability through reparameterization or additional measurements;

4. translates symbolic identifiability results into practical modelling guidance concerning experimental design, parameter estimation, model reformulation, and measurement selection;

5. provides a fully reproducible computational framework within the Julia SciML ecosystem, accompanied by executable Julia notebooks and open-source resources suitable for research and graduate-level instruction.

By integrating mathematical theory, symbolic computation, and reproducible scientific workflows, this tutorial aims to provide a modern reference for researchers seeking to incorporate structural identifiability analysis into mechanistic modelling pipelines.

1.5. Outline of the paper

The remainder of this article is organized as follows. Section 2 introduces the mathematical foundations of structural identifiability, observability, parameter-to-output maps, and identifiable parameter combinations for nonlinear ODE systems. Section 3 presents the computational framework and software workflow implemented in `StructuralIdentifiability.jl`, including the principal symbolic analysis functions and reproducible computational pipeline. Section 4 develops a sequence of representative case studies from epidemiology, pharmacokinetics, and systems biology that illustrate the principal qualitative identifiability phenomena encountered in practice. Section 5 discusses reparameterization strategies and identifiable parameter combinations. Section 6 summarizes best practices for identifiability analysis in applied modelling workflows, while Section 7 addresses reproducible scientific computing considerations. Finally, Sections 8 and 9 provide a broader discussion and concluding remarks.

2. Mathematical Background

2.1. The class of models

We consider parametric ODE systems of the form

$$\Sigma(\theta): \quad \begin{cases} \dot{x}(t) = f(x(t),\, u(t);\, \theta), & x(0) = x_0(\theta), \\ y(t) \;\; = g(x(t),\, u(t);\, \theta), \end{cases} \qquad t \in [0, T], \tag{2.1}$$

where $x(t) \in \mathbb{R}^n$ is the state vector, $u(t) \in \mathbb{R}^m$ a known input, $y(t) \in \mathbb{R}^q$ the measured output, and $\theta \in \Theta \subseteq \mathbb{R}^p$ the parameter vector. The vector fields f and g are assumed to be rational in the state and parameters, an assumption that covers essentially all compartmental, mass action, Michaelis Menten, and viral dynamics models used in practice. Initial conditions may be treated as (a) known constants, (b) additional parameters to be estimated, or (c) generic unknown values; each of these choices alters the identifiability analysis.

2.2. The parameter to output map

Fix the initial conditions and an input $u(\cdot)$. For every $\theta \in \Theta$ such that (2.1) admits a unique solution, the output trajectory is a deterministic function of θ. This defines the parameter to output map

$$\Phi : \Theta \longrightarrow \mathcal{Y}, \qquad \theta \mapsto y(\cdot; \theta), \tag{2.2}$$

where $\mathcal{Y}$ is an appropriate space of output trajectories, for instance $C^\infty([0, T]; \mathbb{R}^q)$. Structural identifiability is a statement about the injectivity of Φ. Geometrically, Φ is the map whose image is what the data can in principle reveal, while its fibres $\Phi^{-1}(\Phi(\theta))$ are precisely the sets of parameter values that the data cannot distinguish.

2.3. Local and global structural identifiability

Following the standard formulation [8, 10, 13], we adopt the following definitions, which hold for generic $\theta \in \Theta$.

Globally identifiable.
A parameter θ_i is globally structurally identifiable if

$$\Phi(\theta) = \Phi(\tilde{\theta}) \implies \theta_i = \tilde{\theta}_i, \quad \text{for almost every } \theta \in \Theta. \tag{2.3}$$

Locally identifiable.
θ_i is locally structurally identifiable if there exists a neighbourhood $U \ni \theta$ such that (2.3) holds on U. Equivalently, $\Phi^{-1}(\Phi(\theta))$ contains only finitely many points sharing the value of θ_i.

Non identifiable.
θ_i is non identifiable if every neighbourhood of θ contains infinitely many values of θ_i consistent with the same output.

The model is globally (respectively locally) identifiable if every θ_i is. Local identifiability is equivalent to the rank fullness of an appropriate sensitivity derivative matrix [22] and is a generic algebraic property. Global identifiability further requires the absence of discrete symmetries, of which the typical examples are the $a \leftrightarrow b$ exchange symmetries in compartmental models and sign symmetries in models with squared rates.

An informal intuition may help. Local identifiability asserts that the output uniquely determines θ once one is close enough; global identifiability asserts that it does so everywhere on Θ. Parameters that are locally but not globally identifiable can be estimated only up to a finite ambiguity that numerical optimisation may resolve arbitrarily depending on initial guesses.

2.4. Observability and the augmented problem

When initial conditions are themselves unknown, identifiability becomes more subtle: one must simultaneously recover θ and x_0. The corresponding concept is observability of the state. We say that $x_i(t)$ is observable if it is determined almost everywhere by $y(\cdot;\theta)$ and the known inputs. Within `StructuralIdentifiability.jl`, declaring an initial condition as known is a substantive piece of information: every observed entry of x_0 contributes algebraic equations that frequently restore identifiability that would otherwise be lost. This effect will be documented repeatedly in the case studies of Section 4.

A natural formalisation is the augmented identifiability problem: regard $(\theta, x_0) \in \Theta \times \mathbb{R}^n$ as a single unknown and apply (2.3) to the joint parameter and initial condition vector. Local and global notions extend immediately. This is the formulation employed by SIAN [17] and `StructuralIdentifiability.jl` whenever initial conditions are not declared known.

2.5. Identifiable functions and reparameterisations

A non identifiable model is not a failed model. The deeper statement is that some functions of the parameters are uniquely determined by the output even when the individual parameters are not. Formally, a rational function $\phi : \Theta \to \mathbb{R}$ is identifiable if

$$\Phi(\theta) = \Phi(\tilde{\theta}) \implies \phi(\theta) = \phi(\tilde{\theta}). \tag{2.4}$$

The collection of identifiable rational functions forms a subfield $\mathbb{K}_{\mathrm{id}} \subseteq \mathbb{Q}(\theta)$. A foundational result of Ovchinnikov, Pillay, Pogudin, and Scanlon [19] establishes that this field is finitely generated and that an explicit set of generators can be computed. These generators are precisely the identifiable combinations of parameters.

Identifiable combinations give an immediate prescription for reparameterisation: replace the original parameter vector by a transformed vector $\eta = (\eta_1, \dots, \eta_r)$, where $\eta_j = \phi_j(\theta)$ are the identifiable generators. In the new coordinates, the model is structurally identifiable by construction. The classical illustration, revisited in Section 5, is the bilinear right hand side $\dot{x} = -pq\,x$, in which only the product pq is identifiable.

2.6. Differential algebra and the input to output relation

The algorithmic backbone underlying global identifiability for nonlinear ODE models is differential algebra [9, 11]. The strategy is to eliminate the unobserved states from (2.1) by repeated differentiation of the output equation, yielding an input to output relation of the form

$$P\big(y, \dot{y}, \ddot{y}, \dots, u, \dot{u}, \dots;\, c(\theta)\big) = 0, \tag{2.5}$$

where P is a polynomial whose coefficients $c(\theta)$ are rational functions of θ. Because (2.5) is uniquely determined, up to a normalisation, by $\Phi(\theta)$, the coefficients $c(\theta)$ are identifiable functions of θ. The question of whether each individual θ_i is recoverable from $c(\theta)$ then reduces to algebraic problems that can be addressed by Gröbner bases, characteristic sets, or, in the implementation adopted by

`StructuralIdentifiability.jl`, by differential elimination through projections [14]. The latter algorithm is markedly faster than classical elimination on realistic biological models and is what renders practical global identifiability analysis of mid sized nonlinear systems tractable in Julia today.

For local identifiability, the considerably lighter Sedoglavic test [22] suffices. It evaluates symbolic sensitivities at a random point and verifies the rank of a Jacobian over a large finite field. It runs in polynomial time in the size of the model and is therefore the appropriate first step in any analysis.

2.7. Summary of the analytic workflow

The mathematical workflow followed throughout the paper is:

1. **Specify the model and measurement scheme.** Write (2.1) explicitly, including the observed states, the known inputs, and the initial conditions that are assumed known.
2. **Local identifiability.** Apply the Sedoglavic test. If a parameter is locally non identifiable, no finer analysis will recover it; it is structurally non identifiable.
3. **Global identifiability.** For parameters that are at least locally identifiable, apply the differential algebra based global test to detect discrete ambiguities.
4. **Identifiable functions.** If non identifiabilities are present, compute the generators of $\mathbb{K}_{\text{id}}$, which point toward the appropriate reparameterisation.
5. **Interpret.** Translate the algebraic verdict into modelling decisions concerning the measurements to add, the parameters to fix, or the groups to estimate jointly.

This is the workflow that `StructuralIdentifiability.jl` operationalises and that the case studies of Section 4 follow without modification. A graphical representation appears in Figure 1.

3. Computational Framework

3.1. The Julia ecosystem for scientific computing

Julia [21] is a high level, just in time compiled programming language designed for technical computing. Three of its design choices are particularly relevant to the present setting. First, Julia generates native code through LLVM and achieves performance comparable to C or Fortran without sacrificing the ergonomics of an interpreted language. Second, multiple dispatch and a rich type system make it natural to write code that is generic in its number type: the same function may operate on floating point numbers, dual numbers for automatic differentiation, or symbolic variables for computer algebra. Third, the package ecosystem is composable: scientific machine learning, ODE solving, parameter estimation, symbolic computation, and identifiability analysis are all designed to interoperate [20].

This composability is what makes `StructuralIdentifiability.jl` substantially more than a stand alone identifiability tool. The same `@ODEmodel` declaration that defines a problem for identifiability analysis can be used to simulate the model with `DifferentialEquations.jl` [20], to fit it with `Turing.jl` or `Optimization.jl`, to compute sensitivities with `ForwardDiff.jl`, or to embed it in a Bayesian workflow. The historical barrier between diagnosing the model and running the model disappears.

3.2. `StructuralIdentifiability.jl`

`StructuralIdentifiability.jl` [14, 19] is a pure Julia package for symbolic structural identifiability analysis of rational ODE models. Its core capabilities are:

- **Local identifiability** of individual parameters and states, computed by a probabilistic, polynomial time algorithm in the size of the model [22].
- **Global identifiability** of individual parameters and states, computed by a probabilistic algorithm with adjustable correctness probability p.
- **Identifiability of user specified rational combinations** of parameters.
- **Computation of generators of all identifiable functions** of the parameters (and optionally of the states as well).
- **Composability** with the broader SciML ecosystem.

The package supports rational right hand sides, multiple inputs, multiple outputs, and arbitrary combinations of known and unknown initial conditions.

3.3. The core API

The four constructs used throughout the case studies are summarised below.

3.3.1. `@ODEmodel`

The `@ODEmodel` macro provides a near mathematical syntax for declaring a system:

```
using StructuralIdentifiability

ode = @ODEmodel(
    x1'(t) = -k * x1(t),                    # state equations
    x2'(t) =  k * x1(t) - g * x2(t),
    y(t)   =  x2(t)                         # observed output(s)
)
```

Every symbol on a left hand side that is differentiated is a state. Every other symbol on the right hand sides is automatically classified as a parameter unless it is declared as an input by appearing as `u(t)`. Every symbol on the left of an output equation `y...(t) = ...` is an observed output. Multiple outputs and multiple inputs are supported by listing them as separate equations.

3.3.2. `assess_local_identifiability`

The function `assess_local_identifiability(ode)` executes Sedoglavic's probabilistic algorithm [22] and returns, for each parameter and each state at time 0, either the symbol `:locally` or `:nonidentifiable`. A keyword argument `p` sets the desired probability of correctness, with a default value of 0.99 that is sufficient for most use cases. Because the test only checks the rank of a sensitivity matrix at a generic point, it is fast, typically completing in milliseconds for the models considered in this paper, and constitutes the appropriate first call in any analysis.

3.3.3. `assess_identifiability`

The function `assess_identifiability(ode)` returns the global verdict. Each parameter and each state is labelled `:globally`, `:locally` (i.e., locally but not globally), or `:nonidentifiable`. By default the analysis is performed on every individual parameter. A list of rational combinations can be passed as the keyword argument `funcs_to_check` to test specific combinations such as `funcs_to_check = [a01 + a12, a01 * a12]`. The underlying algorithm uses differential elimination through projections [14]; its complexity scales with the number of states and the differential order required to eliminate them.

3.3.4. `find_identifiable_functions`

The function `find_identifiable_functions(ode)` returns a generating set of $\mathbb{K}_{\text{id}}$. This is a finite list of rational functions of the parameters whose values together determine every other identifiable function [19]. With the keyword argument `with_states = true`, the procedure additionally generates the field of observable functions of parameters and states. This is the natural object when initial conditions are unknown.

Conceptually, `find_identifiable_functions` is the constructive counterpart to `assess_identifiability`. The latter labels parameters; the former tells the modeller what should be estimated in their place.

3.4. The computational pipeline

The notebook accompanying this article packages the foregoing calls into the uniform pipeline summarised in Table 1 and applies it to every case study. Figure 1 provides a schematic representation.

Table 1: Pseudocode of the recommended structural identifiability pipeline. Each numbered step corresponds to a single call into `StructuralIdentifiability.jl`, executed from the same Julia session in which the model is defined. The pipeline is applied without modification to every case study in Section 4.

Step	Julia call	Purpose
1	`ode = @ODEmodel(...)`	Declare states, parameters, inputs, outputs, and any known initial conditions.
2	`assess_local_identifiability(ode)`	Run the fast Sedoglavic rank test; halt and reformulate if numerous parameters are flagged as non identifiable.
3	`assess_identifiability(ode)`	Run the global test (differential elimination); detect parameters that are locally but not globally identifiable.
4	`find_identifiable_functions(ode)`	Compute the generators of $\mathbb{K}_{\text{id}}$ whenever non identifiabilities are present; obtain the appropriate parameter vector for estimation.
5	(repeat 2–4 with `ic` known)	Optionally repeat the analysis assuming selected initial conditions are known and compare verdicts.
6	(interpret)	Translate the algebraic verdict into modelling decisions: which measurements to add, which parameters to fix, which combinations to report.

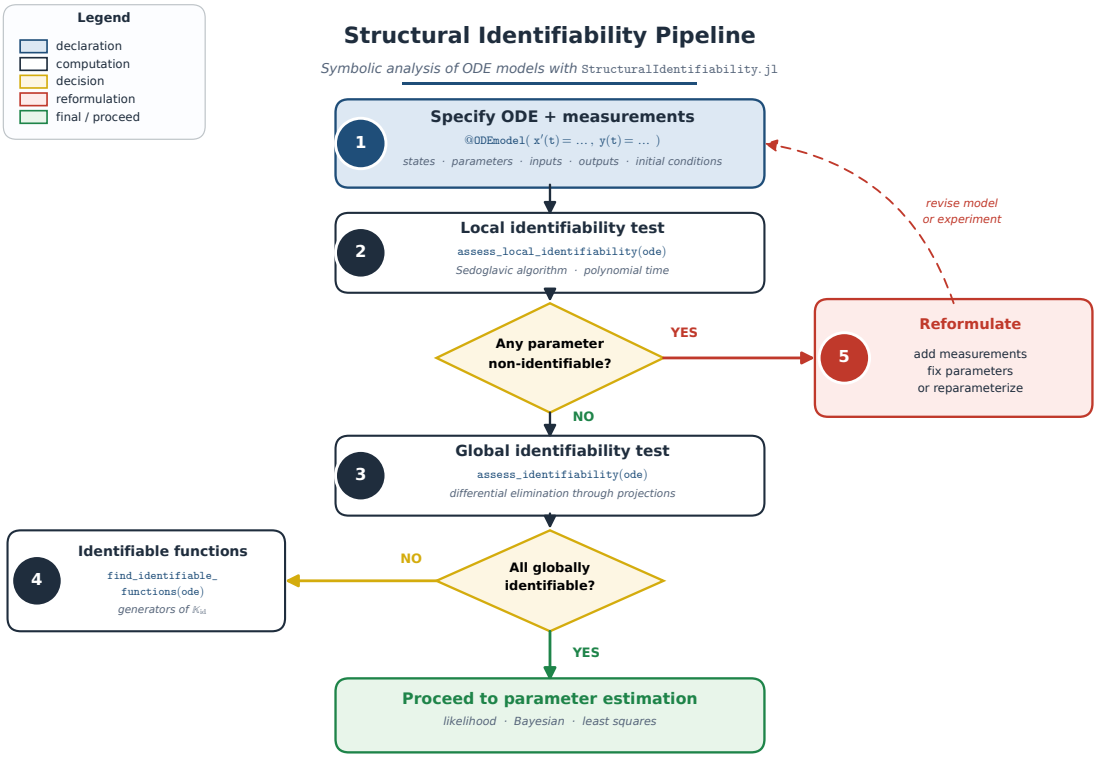


Figure 1: Workflow diagram of the structural identifiability pipeline. Beginning from a model declared via the `@ODEmodel` macro, a fast local identifiability test (Sedoglavic algorithm) is applied. If any parameter is diagnosed as non identifiable, the modeller is directed to the reformulation step (additional measurements, fixed parameters, or reparameterisation). Otherwise the global identifiability test (differential elimination through projections) refines the verdict. Parameters reported as locally but not globally identifiable, or as non identifiable, are routed to `find_identifiable_functions`, which returns the generators of the identifiable field $\mathbb{K}_{\text{id}}$ that constitute the appropriate parameter vector for estimation. The entire pipeline runs in the same Julia session in which the model is defined and simulated.

4. Case Studies

The case studies that follow are arranged in approximately increasing complexity. Each one introduces a distinct conceptual lesson: how the analysis behaves on a trivially identifiable model, how it detects local only identifiability, how the addition of a measurement restores identifiability, how reparameterisation works in practice, and how systems biology style structural issues manifest within compartmental epidemiology. A schematic overview of all six mechanistic models is given in Figure 2, with the observed compartments highlighted in each panel.

For every model we follow a uniform template comprising scientific background, ODE system, parameters and variables, observed outputs, identifiability results obtained from `StructuralIdentifiability.jl`, and a discussion of the implications for parameter estimation and experimental design. A consolidated summary of the verdicts appears in Table 2 and Figure 3.

4.1. Exponential decay: a minimal benchmark

Background. The exponential decay model is the simplest non trivial ODE encountered in pharmacokinetics (single compartment elimination), radioactive decay, first order chemical kinetics, and many related settings. It serves here as a calibration example: a model that is globally identifiable for the most evident reason possible.

Model. Let $x(t)$ denote a concentration or population and $k > 0$ the first order decay rate:

$$\dot{x}(t) = -k\,x(t), \qquad x(0) = x_0, \qquad y(t) = x(t). \tag{4.1}$$

The parameter vector is $\theta = (k)$ and the state is observed directly.

Julia specification.

```
ode = @ODEmodel(
    x'(t) = -k * x(t),
    y(t)  =  x(t)
)
```

Identifiability results. Both `assess_local_identifiability` and `assess_identifiability` return `k => :globally` and `x(t) => :globally`. The closed form solution $x(t) = x_0 e^{-kt}$ renders this transparent: a single noiseless observation at any positive time, combined with $x(0)$, uniquely determines k. When $x(0)$ is unknown but $x(t)$ is observed continuously, any two distinct time points suffice.

Implication. The exponential model serves as a sanity check. Any analysis pipeline should certify it as globally identifiable, and `StructuralIdentifiability.jl` does so essentially instantaneously. The example also encapsulates the most elementary structural fact: identifiability is decided by the geometry of the parameter to output map, not by the quantity of data.

4.2. The SIR model: a classical compartmental example

Background. The Susceptible, Infectious, Recovered (SIR) model is the founding mechanistic model of mathematical epidemiology [24]. It captures the qualitative dynamics of an outbreak through two rate parameters: the transmission rate β and the recovery rate γ. In surveillance practice, the closest empirically available quantity is incidence, namely the rate of new infections per unit time, which corresponds to βSI.

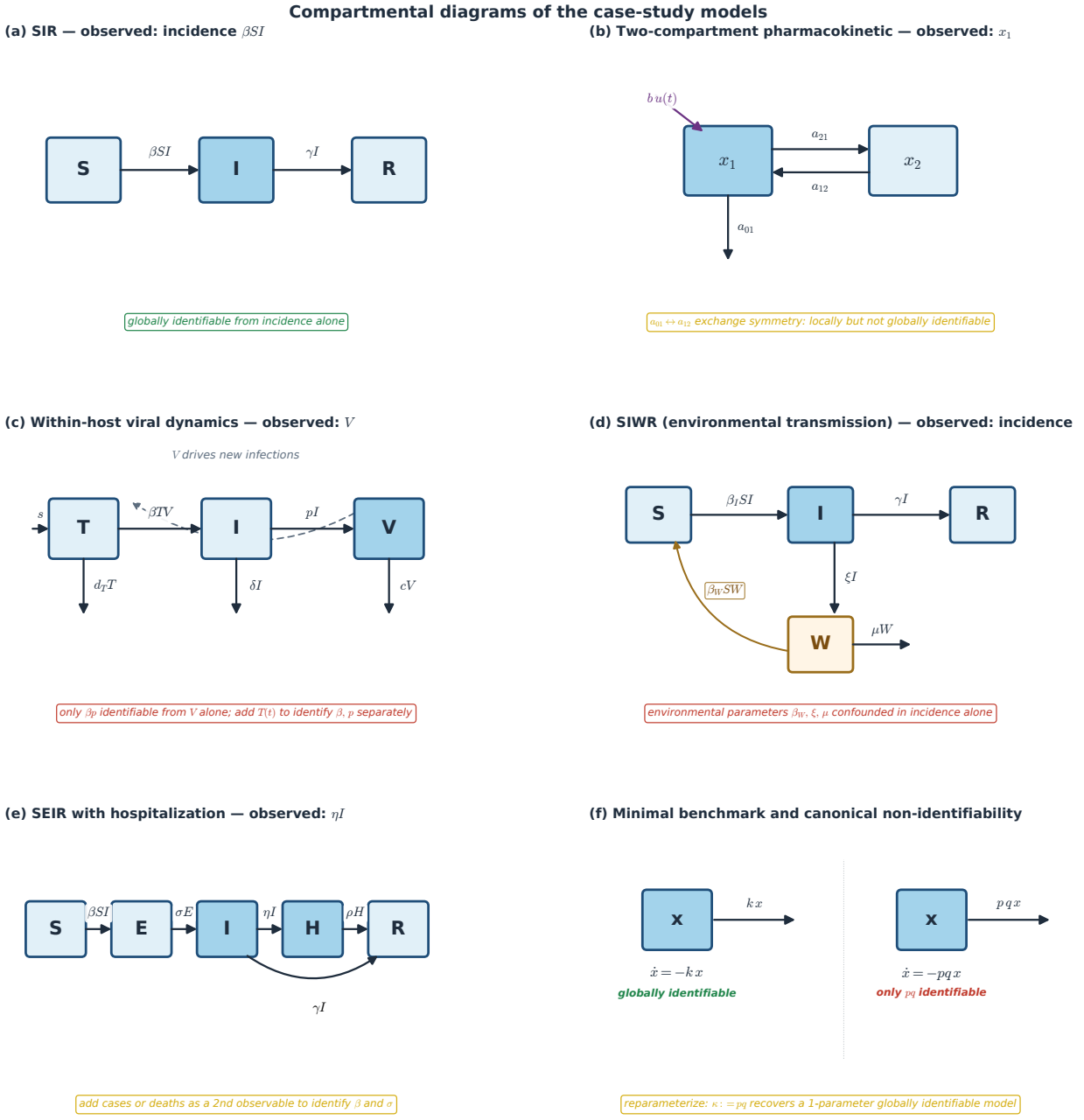


Figure 2: Compartmental diagrams of the mechanistic models analysed in Section 4. Each panel shows the state variables, the parameter labelled flows, and the observed compartments (shaded). (a) SIR model with incidence βSI as the observed output. (b) Two compartment pharmacokinetic model with the central compartment x_1 observed; the $a_{01} \leftrightarrow a_{12}$ exchange symmetry renders these two parameters locally but not globally identifiable. (c) Within host viral dynamics in the spirit of Perelson, with free virus V observed; only the product βp is identifiable from V alone. (d) SIWR model with environmental reservoir W; aggregated incidence cannot disentangle the environmental parameters β_W, ξ, μ. (e) SEIR model with hospitalisation H; hospitalisation incidence ηI alone is insufficient to identify β and σ. (f) Exponential decay (globally identifiable) and bilinear toy model (only pq identifiable), serving as a minimal benchmark and a canonical non identifiability example, respectively.

Model. With state vector (S, I, R) and constant total population, the model takes the form

$$\begin{aligned} \dot{S} &= -\beta\, SI, \\ \dot{I} &= \beta\, SI - \gamma\, I, \\ \dot{R} &= \gamma\, I, \end{aligned} \tag{4.2}$$

with $\theta = (\beta, \gamma)$. The natural observed output is the incidence

$$y(t) = \beta\, S(t)\, I(t). \tag{4.3}$$

Julia specification.

```
ode = @ODEmodel(
    S'(t) = -beta * S(t) * I(t),
    I'(t) =  beta * S(t) * I(t) - gamma * I(t),
    R'(t) =  gamma * I(t),
    y(t)  =  beta * S(t) * I(t)
)
```

Identifiability results. `assess_identifiability` returns $\beta \mapsto$ `:globally` and $\gamma \mapsto$ `:globally`, with the state R correctly reported as non observable since R does not appear in any input to output relation derived from y. The basic reproduction number $\mathcal{R}_0 = \beta S_0/\gamma$ is therefore identifiable whenever S_0 is known.

Implication. The SIR model offers a useful counterexample to a common misconception: incidence alone is sufficient to identify both rate parameters of the model, despite the fact that the cumulative compartment R is not observed. The reason is algebraic. Differentiating $y(t) = \beta SI$ once and substituting the state equations produces an input to output relation that involves only β, γ, y, $\dot{y}$, and known constants. This is precisely the relation that `StructuralIdentifiability.jl` computes internally [14].

4.3. A two compartment pharmacokinetic model

Background. Linear compartmental models constitute the workhorse of pharmacokinetics: a drug moves between a central blood compartment and one or more peripheral tissues, is eliminated through one or more clearance pathways, and is typically measured in plasma. The classical two compartment model has been a canonical case study for identifiability since Bellman and Åström [5], and it is the textbook example of a model with locally but not globally identifiable parameters and an exchange symmetry [7, 8].

Model. Let x_1 denote the central (sampled) compartment and x_2 a peripheral compartment. Let a_{12} and a_{21} be the inter compartmental rate constants, a_{01} the elimination rate from x_1, and b a bolus input gain:

$$\begin{aligned} \dot{x}_1 &= -(a_{01} + a_{21})\, x_1 + a_{12}\, x_2 + b\, u, \\ \dot{x}_2 &= a_{21}\, x_1 - a_{12}\, x_2, \\ y &= x_1. \end{aligned} \tag{4.4}$$

The parameter vector is $\theta = (a_{01}, a_{12}, a_{21}, b)$.

Julia specification.

```
ode = @ODEmodel(
    x1'(t) = -(a01 + a21) * x1(t) + a12 * x2(t) + b * u(t),
    x2'(t) =   a21 * x1(t) - a12 * x2(t),
    y(t)   =   x1(t)
)
```

Identifiability results. `assess_identifiability(ode)` yields, in the typical output produced by the package:

```
x1(t)  => :globally
x2(t)  => :nonidentifiable
a01    => :locally
a12    => :locally
a21    => :globally
b      => :nonidentifiable
```

Two non trivial structural facts are exposed simultaneously. The pair (a_{01}, a_{12}) is locally identifiable but globally ambiguous, since the model is invariant under the permutation that exchanges a_{01} and a_{12} while preserving their sum $a_{01} + a_{12}$ and product $a_{01}\, a_{12}$. The input gain b and the unobserved compartment x_2 are non identifiable because only the product of the gain with the magnitude of x_2 enters the output.

Invoking

```
assess_identifiability(ode, funcs_to_check = [a01 + a12, a01 * a12])
```

confirms that both $a_{01}+a_{12}$ and $a_{01}\, a_{12}$ are globally identifiable, while `find_identifiable_functions(ode)` returns a generating set that includes these symmetric combinations together with a_{21} [19, 16].

Implication. This is the prototype of a model in which parameter estimation, as posed, is ill defined, yet a meaningful and unique inference remains possible by estimating symmetric combinations of the rate constants. Practically, this means reporting $(a_{01} + a_{12},\ a_{01}\, a_{12},\ a_{21})$ rather than the four original constants, or fixing one rate from a separate experiment such as a single compartment study. The example is also a cautionary tale. Numerical optimisation on (4.4) will return one of two parameter triples that are pharmacologically distinct, corresponding to different elimination versus distribution interpretations, but generate identical plasma curves. This phenomenon is demonstrated empirically in Figure 4(B).

4.4. Within host viral dynamics in the spirit of Perelson

Background. The within host viral dynamics model of Perelson and colleagues [25] revolutionised HIV modelling by reducing complex immune dynamics to three coupled compartments: uninfected target cells T, infected cells I, and free virus V. The framework quantified rate parameters that had previously appeared inaccessible. Variants of the model are now used routinely for HIV, hepatitis B and C, SARS-CoV-2, and numerous other infections. Identifiability of these models is a recurrent question, both in the original work and in the subsequent literature [10].

Model. A standard formulation reads

$$\begin{aligned} \dot{T} &= s - d_T\, T - \beta\, T V, \\ \dot{I} &= \beta\, T V - \delta\, I, \\ \dot{V} &= p\, I - c\, V, \end{aligned} \tag{4.5}$$

with $\theta = (s, d_T, \beta, \delta, p, c)$. Free virus V is the only directly measurable quantity in most clinical studies,

$$y(t) = V(t). \tag{4.6}$$

Julia specification.

```
ode = @ODEmodel(
    T'(t) = s - d_T * T(t) - beta * T(t) * V(t),
    I'(t) = beta * T(t) * V(t) - delta * I(t),
    V'(t) = p * I(t) - c * V(t),
    y(t)  = V(t)
)
```

Identifiability results. When only V is measured, `assess_identifiability` typically returns the rates δ and c as globally identifiable, while β and p appear only through the product βp. The state $T(0)$ and the source term s are coupled and not individually identifiable. The reason is that the virus equation $\dot{V} = pI - cV$ sees infected cells only through the product pI, and the infected cell production term $\dot{I} = \beta TV - \delta I$ sees the target cell population only through the product βTV. Tracing the chain, only the combined burst rate quantity βp leaves a trace in $V(t)$, recovering the well known result of Miao et al. [10].

The call

```
find_identifiable_functions(ode)
```

returns generators that include δ, c, and the combination βp, together with auxiliary functions of s and d_T.

Restoring identifiability. If the experiment additionally measures the uninfected target cell count, $y_2(t) = T(t)$, corresponding to flow cytometry in HIV studies, then β and p become individually identifiable. Re executing `assess_identifiability` after augmenting the observation confirms this: each of the six original parameters becomes globally identifiable.

Implication. The Perelson style example is the prototypical case of non identifiability remedied by additional measurement. Modellers fitting only viral load are estimating βp, regardless of how the optimiser reports its results, and any biological interpretation of β or p in isolation is unwarranted. The example further illustrates why identifiability analysis must be performed in the context of the specific measurement scheme; the model and the experiment are inseparable [2].

4.5. Environmentally transmitted disease: an SIWR model

Background. For pathogens with substantial environmental transmission, such as *Vibrio cholerae*, rotavirus, norovirus, and certain respiratory pathogens, the standard SIR framework must be augmented with a pathogen reservoir compartment W representing water or the environment. The

resulting SIWR model has been a focal example in epidemiological identifiability literature [26], because the environmental compartment alters the structure of the inverse problem in a non trivial way.

Model.

$$\begin{aligned} \dot{S} &= -\beta_I\, SI - \beta_W\, SW, \\ \dot{I} &= \beta_I\, SI + \beta_W\, SW - \gamma\, I, \\ \dot{W} &= \xi\, I - \mu\, W, \\ \dot{R} &= \gamma\, I, \end{aligned} \tag{4.7}$$

where β_I denotes direct (person to person) transmission, β_W environmental transmission, ξ the pathogen shedding rate into the environment, μ the environmental decay rate, and γ the recovery rate. Surveillance typically yields the aggregated incidence

$$y(t) = \beta_I\, S(t)I(t) + \beta_W\, S(t)W(t). \tag{4.8}$$

Julia specification.

```
ode = @ODEmodel(
    S'(t) = -beta_I*S(t)*I(t) - beta_W*S(t)*W(t),
    I'(t) =  beta_I*S(t)*I(t) + beta_W*S(t)*W(t) - gamma*I(t),
    W'(t) =  xi*I(t) - mu*W(t),
    R'(t) =  gamma*I(t),
    y(t)  =  beta_I*S(t)*I(t) + beta_W*S(t)*W(t)
)
```

Identifiability results. When only the aggregated incidence is observed, the rate γ and one of the two transmission routes are typically identifiable, while the environmental parameters β_W, ξ, and μ enter only through identifiable combinations such as $\beta_W\,\xi$ and β_W/μ. The output cannot distinguish a weakly environmental but heavily shedding pathogen from a strongly environmental but lightly shedding pathogen, because the two scenarios are observationally equivalent in the incidence signal [26].

`find_identifiable_functions` makes this explicit by returning generators that involve products and ratios of the environmental parameters rather than the parameters themselves. Adding a direct measurement of the environmental compartment, $y_2(t) = W(t)$, corresponding to an environmental pathogen surveillance programme, separates the products and renders all parameters individually identifiable.

Implication. The SIWR example is operationally important: it illustrates how identifiability analysis can inform experimental design in public health. The fact that aggregated incidence confounds environmental and shedding parameters is precisely the kind of result that motivates investment in environmental sampling, such as water testing or wastewater surveillance, as part of cholera or hepatitis A response.

4.6. An SEIR model with hospitalisation

Background. During the COVID-19 pandemic, hospitalisation tracking models in the SEIR-H family became the dominant format for short term forecasts. A latent exposed compartment E captures incubation; hospitalisation rates inform health system capacity planning; recovery rates

inform the overall epidemic timescale. The natural observations are hospitalisation incidence and, where available, reported case counts [23].

Model.

$$\begin{aligned} \dot{S} &= -\beta\, SI, \\ \dot{E} &= \beta\, SI - \sigma\, E, \\ \dot{I} &= \sigma\, E - (\gamma + \eta)\, I, \\ \dot{H} &= \eta\, I - \rho\, H, \\ \dot{R} &= \gamma\, I + \rho\, H, \end{aligned} \tag{4.9}$$

where σ is the rate of progression from exposed to infectious, η the hospitalisation rate, ρ the rate of hospital discharge, and γ the recovery rate for non hospitalised individuals. Hospitalisation incidence $y_1 = \eta I$ is the primary observation, augmented optionally by case incidence $y_2 = \beta SI$.

Julia specification.

```
ode = @ODEmodel(
    S'(t) = -beta*S(t)*I(t),
    E'(t) =  beta*S(t)*I(t) - sigma*E(t),
    I'(t) =  sigma*E(t) - (gamma + eta)*I(t),
    H'(t) =  eta*I(t) - rho*H(t),
    R'(t) =  gamma*I(t) + rho*H(t),
    y1(t) =  eta*I(t)
)
```

Identifiability results. With only $y_1 = \eta I$ observed, `assess_identifiability` typically reports the hospitalisation rate η, the discharge rate ρ, and the combination $\gamma + \eta$ as identifiable, while σ and β enter only through products with the unobservable initial exposed population. Adding case incidence $y_2 = \beta SI$ (or, equivalently, σE) restores the individual identifiability of σ and β.

Variants of this model that incorporate disease induced mortality have been analysed in detail in the literature; the addition of a second observable corresponding to the daily death count consistently restores global identifiability of the full parameter vector [23].

Implication. The SEIR-H example illustrates that even widely deployed forecasting models silently rely on multiple data streams for identifiability. Reporting a single data stream while estimating five or six parameters constitutes a structural impossibility; in practice the resulting estimates are determined by priors. A formal identifiability analysis prior to model deployment would expose this circumstance and motivate either the inclusion of additional data or a reduced parameterisation.

4.7. Consolidated verdicts

The identifiability verdicts established across all six mechanistic models are summarised in Table 2 and represented graphically in Figure 3. Two structural lessons emerge immediately. First, the verdict for a given parameter is not a property of the model alone; it depends jointly on the model and on the measurement scheme. This is visible in the table as the contrast between the rows corresponding to a restricted output (V only, incidence only, ηI only) and those corresponding to an augmented output. Second, non identifiability is rarely total. In every case considered here,

the restricted output schemes that fail to identify all parameters nonetheless identify a non trivial subset, sometimes after the parameters have been reorganised into identifiable combinations.

Table 2: Consolidated summary of the case studies analysed in Section 4. For each model and measurement scheme, the table lists the number of states, the number of free parameters, the observed output, the global identifiability verdict, an indicative set of identifiable functions returned by `find_identifiable_functions`, and the section in which the analysis is presented. Verdict abbreviations: **G** globally identifiable; **L** locally but not globally; **NI** non identifiable; *partial* indicates that at least one parameter falls into each of two categories.

Model	States	Params	Observed output	Verdict	Identifiable functions (illustrative)	§
Exponential decay	1	1	$y = x$	G	$\{k\}$	4.1
SIR	3	2	$y = \beta SI$	G	$\{\beta, \gamma\}$	4.2
Two compartment PK	2	4	$y = x_1$	partial (L/NI)	$\{a_{21},\ a_{01} + a_{12},\ a_{01}\, a_{12}\}$	4.3
Viral dynamics	3	6	$y = V$	partial (G/NI)	$\{\delta,\ c,\ \beta p,\ \dots\}$	4.4
Viral dynamics, augmented	3	6	$y = (V, T)$	G	$\{s,\ d_T,\ \beta,\ \delta,\ p,\ c\}$	4.4
SIWR (incidence)	4	5	$y =$ incidence	partial	$\{\beta_I,\ \gamma,\ \beta_W\, \xi,\ \beta_W/\mu\}$	4.5
SIWR (incidence+W)	4	5	$y =$ (incidence, W)	G	$\{\beta_I,\ \beta_W,\ \xi,\ \mu,\ \gamma\}$	4.5
SEIR-H (ηI)	5	5	$y = \eta I$	partial	$\{\eta,\ \rho,\ \gamma + \eta,\ \dots\}$	4.6
SEIR-H ($\eta I + \beta SI$)	5	5	$y = (\eta I,\ \beta SI)$	G	$\{\beta,\ \sigma,\ \gamma,\ \eta,\ \rho\}$	4.6
Bilinear toy	1	2	$y = x$	NI	$\{pq\}$	5

5. Reparameterisation and Identifiable Combinations

5.1. The canonical bilinear example

We pause the procession of realistic models to examine, with maximal clarity, the simplest non trivial structural non identifiability. Consider the bilinear scalar system

$$\dot{x}(t) = -p\, q\, x(t), \qquad y(t) = x(t), \tag{5.1}$$

with $\theta = (p, q)$ and $x(0)$ known. The closed form solution $x(t) = x(0)\, e^{-pqt}$ depends on p and q only through their product. The pair $(p, q) = (2, 3)$ and the pair $(p, q) = (6, 1)$ generate identical output trajectories; no experiment can distinguish them. Figure 4(A) provides a direct numerical verification: four distinct (p, q) pairs sharing $pq = 6$ collapse onto a single output curve.

`StructuralIdentifiability.jl` confirms the algebraic statement immediately:

```
ode = @ODEmodel(
    x'(t) = -p * q * x(t),
    y(t)  =  x(t)
)
assess_identifiability(ode)
# p => :nonidentifiable
# q => :nonidentifiable
find_identifiable_functions(ode)
# returns [p*q]
```

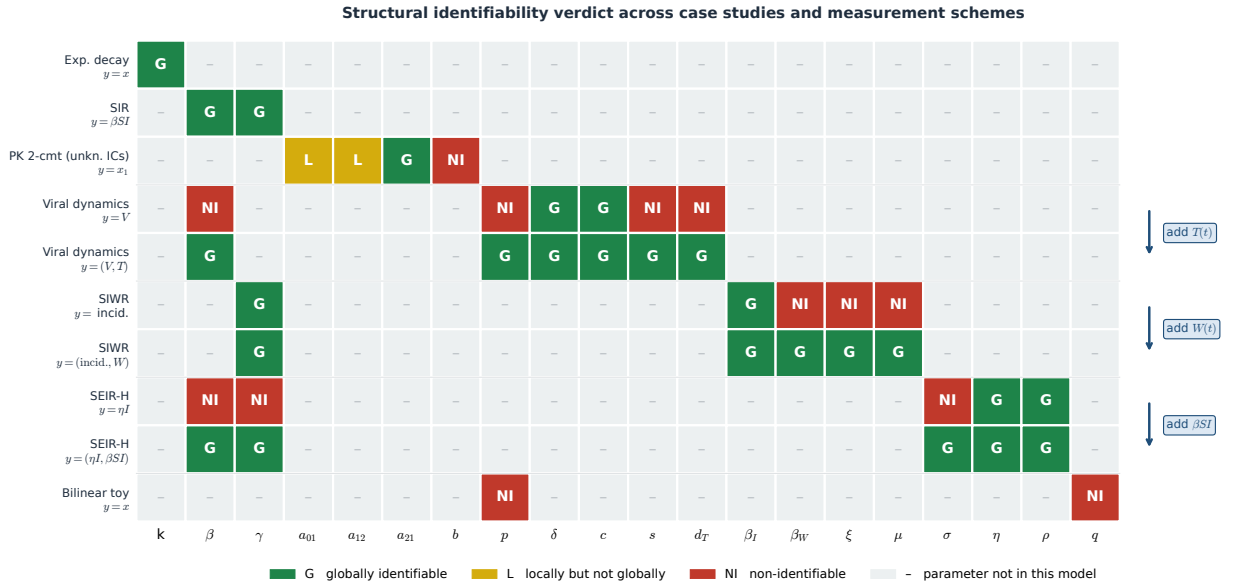


Figure 3: Structural identifiability verdict matrix across all case studies and measurement schemes. Rows correspond to a model and measurement scheme; columns correspond to individual parameters. Cells are colour coded as globally identifiable (**G**, green), locally but not globally identifiable (**L**, amber), or non identifiable (**NI**, red); grey cells indicate parameters not present in the relevant model. The matrix exposes the central pedagogical point of the paper: augmenting the measurement scheme can flip entire rows from red to green. Concretely, adding $T(t)$ to viral dynamics, $W(t)$ to SIWR, or βSI to SEIR-H restores global identifiability of every parameter in the respective model.

The model is non identifiable in p and q, yet the product pq is globally identifiable. The reparameterisation is therefore canonical: introducing $\kappa := pq$ yields

$$\dot{x}(t) = -\kappa\, x(t),$$

which is precisely the exponential decay model of Section 4.1. The reformulated model contains one parameter, is globally identifiable, and captures exactly the information that the data can deliver.

The analogous phenomenon for the two compartment pharmacokinetic model is illustrated in Figure 4(B). Two parameter triples that exchange the values of a_{01} and a_{12} while preserving their sum and product produce trajectories that are indistinguishable to numerical precision. The verdict of Section 4.3, namely that a_{01} and a_{12} are locally but not globally identifiable, is thus borne out by direct simulation.

5.2. Identifiable combinations as a constructive object

The bilinear example crystallises a general principle. Whenever `find_identifiable_functions` returns generators that are non trivial functions of the original parameters, those generators constitute the model's appropriate parameter vector. Three immediate consequences follow.

1. **Fitting.** Likelihood, Bayesian, and least squares estimation should be performed in the new coordinates $\eta = \phi(\theta)$ rather than in θ. Optimisation on θ is ill posed, whereas optimisation on η is well posed by construction.

2. **Reporting.** Confidence intervals and posterior credible regions should be reported on η.

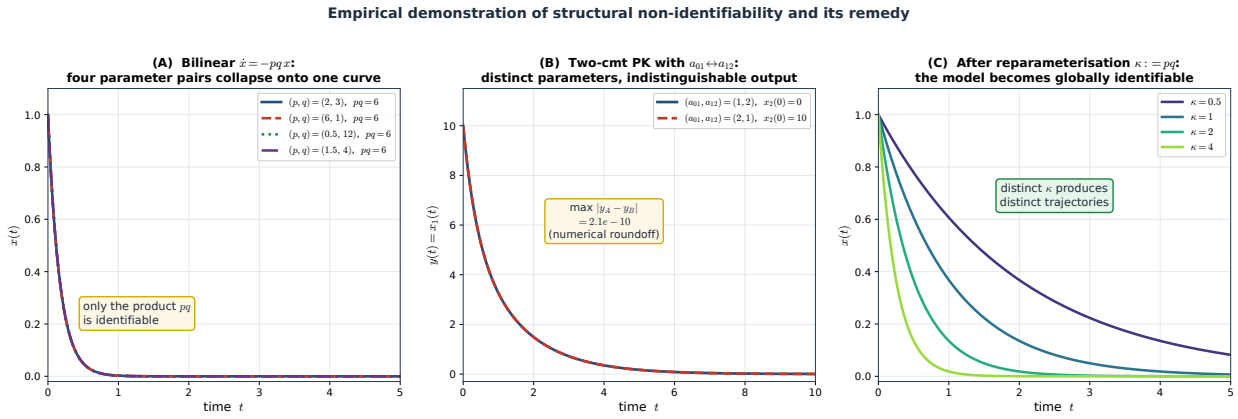


Figure 4: Empirical demonstration of structural non identifiability through ODE integration at high tolerance. (A) Bilinear toy model $\dot{x} = -pq\,x$ integrated for four distinct pairs $(p, q) = (2, 3), (6, 1), (0.5, 12), (1.5, 4)$, all sharing the product $pq = 6$. The four trajectories overlay perfectly, illustrating that only the product is identifiable. (B) Two compartment pharmacokinetic model of Section 4.3 integrated for two parameter sets related by the $a_{01} \leftrightarrow a_{12}$ exchange: $(a_{01}, a_{12}, a_{21}) = (1, 2, 0.5)$ with $x_2(0) = 0$, and $(a_{01}, a_{12}, a_{21}) = (2, 1, 0.5)$ with $x_2(0) = 10$, both having $x_1(0) = 10$. The two trajectories agree to within roundoff error (maximum absolute difference 2.1×10^{-10}), confirming that the parameters a_{01} and a_{12} are locally but not globally identifiable.

Intervals on θ in the absence of identifiability are not interpretable; they reflect the prior or the regulariser [1].

3. **Comparing studies.** Two studies that independently estimate θ from non identifiable models cannot be combined meaningfully, whereas two studies that estimate η can.

5.3. When reparameterisation is not enough

There exist settings in which the identifiable functions are scientifically uninterpretable, for instance when a sum of dissimilar rate constants whose individual values are biologically meaningful turns out to be the only identifiable combination. In such cases the available remedies are:

- **Add measurements.** If the identifiable field is too small, additional outputs (a second compartment, an environmental measurement, or a second experimental condition) enlarge it.
- **Fix parameters from external data.** If a parameter is known from an independent experiment, it can be removed from the parameter vector, and the remaining parameters may become identifiable.
- **Add inputs.** A controllable input $u(t)$ can substantially enlarge the identifiable field. This is the principle behind input design in pharmacokinetics [8].

The unifying theme is that identifiability is a property of both the model and the experiment, and the modeller has leverage on both.

6. Best Practices and Recommendations

We collect, in approximate order of importance, the practical recommendations that emerge from the case studies and from extended experience with these tools.

6.1. Test identifiability before parameter estimation

The most damaging failure mode in applied modelling is to fit a non identifiable model to data, observe a numerical optimum, and report point estimates accompanied by ostensibly plausible confidence intervals. The optimum is, in reality, one element of a continuum; the confidence interval reflects the regulariser; the point estimate is an artefact [1, 2]. The cost of executing `assess_local_identifiability` is essentially zero, and doing so should be as standard as verifying solver tolerances.

6.2. Start local, then go global

For models of moderate size, run `assess_local_identifiability` first. If it certifies global identifiability, `assess_identifiability` will refine the verdict to global versus local only. If it diagnoses non identifiability, no global tool will help; the analysis terminates there and reformulation begins.

6.3. Treat the measurement scheme as part of the model

Identifiability depends jointly on f and g. Models that are non identifiable for one output are frequently globally identifiable for another. The case studies of Sections 4.4 and 4.5 make this explicit: adding a measurement of $T(t)$ or $W(t)$ converts a non identifiable system into a globally identifiable one. The first design question for any inverse problem should therefore be: which outputs make this model identifiable, and which of those outputs are practical to collect?

6.4. Be precise about initial conditions

Whether x_0 is treated as known, parameterised, or generic can overturn the identifiability verdict for half of the parameters in a typical epidemiological model. The case studies systematically perform two analyses, one with unknown and one with known initial conditions, and report both [23]. This is good practice in general.

6.5. Use identifiable combinations as the reporting unit

If `find_identifiable_functions` returns combinations, report them. A paper claiming to have estimated β and p in a viral dynamics model from viral load alone is, with high probability, reporting an artefact; a paper that estimates the combination βp is reporting a fact [10].

6.6. Distinguish structural from practical identifiability

Structural identifiability is a necessary but not sufficient condition for reliable parameter estimation. A structurally identifiable parameter can still be practically non identifiable when the sensitivity of the output to that parameter is small relative to measurement noise [1, 2]. Once structural identifiability has been established, the appropriate subsequent steps are profile likelihoods, the Fisher Information Matrix, and Markov chain Monte Carlo diagnostics, all of which interrogate the data side of the inverse problem.

6.7. Common mistakes to avoid

The recurring pitfalls are:

- estimating a model from only cumulative observations when only incidence is informative, or vice versa;
- treating exchange symmetric parameters as if they were uniquely estimated by an optimiser [7];
- fixing initial conditions to convenient round numbers without acknowledging the structural commitment that this entails;
- interpreting non identifiability as a software bug rather than as a model property;
- failing to repeat the identifiability analysis after the model is modified.

6.8. Limitations of structural identifiability

Structural identifiability is silent on three issues that matter in practice:

- It does not account for noise. A parameter may be globally identifiable in theory but estimable only to a wide credible interval given realistic data [1].
- It does not address model misspecification. A globally identifiable but incorrectly specified model is still incorrectly specified.
- It is restricted to rational models. Systems with non rational nonlinearities, including piecewise defined feedback and neural network augmented dynamics, require approximation or alternative numerical approaches.

These are limitations to be aware of, not reasons to skip the analysis.

6.9. Future directions

Algorithmic frontiers include identifiability for delay differential equations, stochastic ODE models, partially observed partial differential equations, and agent based models. The Julia ecosystem is well positioned to address these directions, and future work integrating `StructuralIdentifiability.jl` with the broader SciML stack on inference, automatic differentiation, and neural ODE models constitutes a particularly promising research line.

7. Reproducibility and Open Science

7.1. The reproducibility argument

A scientific computation that cannot be reproduced remains in epistemic limbo: it is a claim, not yet a result. Identifiability analysis is no exception. The reproducibility of a structural identifiability claim, namely that a given model is globally identifiable under a given measurement scheme, depends on the exact model specification, the exact software version, and the random seeds of any probabilistic algorithm. Anyone wishing to build on, contest, or extend the claim must be able to rerun the analysis. Three practical commitments make this realistic.

7.2. Open source tooling

`StructuralIdentifiability.jl`, Julia [21], and the broader SciML stack [20] are released under permissive open source licences. This eliminates the institutional and financial barriers that have constrained the use of closed commercial computer algebra tools, and makes the analysis runnable

in any environment, including continuous integration pipelines, cloud workspaces, classroom servers, and personal laptops.

7.3. Notebook based workflows

The companion notebook for this article is a Jupyter notebook with a Julia kernel. Every command, every output, and every model resides in the notebook in the order one would naturally execute them. The notebook is the document and the document is the experiment. Pinning package versions through a paired `Project.toml` and `Manifest.toml` guarantees that the same notebook will produce identical results indefinitely.

7.4. Archival deposition

To support reproducibility and long-term accessibility, all Julia notebooks, source code, figures, and computational dependencies used in this tutorial have been archived on Zenodo with a citable DOI [3]. The archived repository provides persistent versioning, long-term preservation, and a fully reproducible record of the computational workflow developed throughout this study. Public archival of computational resources is strongly recommended for applied identifiability analyses, as it promotes transparency, reproducibility, and reuse within the broader scientific community.

7.5. Reproducibility as pedagogy

A final, often underappreciated benefit is pedagogical. A reproducible identifiability notebook is a teaching artefact of unusual quality: students who load it, modify a measurement scheme, and observe the verdict change in real time acquire an intuition about identifiability that is otherwise difficult to develop. This style of instruction has proved particularly effective in graduate courses in mathematical biology, mathematical epidemiology, and computational systems biology.

8. Discussion

Structural identifiability analysis addresses a fundamental question in mechanistic modelling: whether the parameters of a dynamical system can be uniquely recovered from the available observations under a given measurement scheme. Although often viewed as a theoretical topic, identifiability is directly connected to the reliability of parameter estimation, predictive modelling, uncertainty quantification, and mechanistic interpretation. The case studies presented throughout this tutorial demonstrate that non-identifiability is neither rare nor restricted to highly complex systems. As summarized in Table 2 and Figure 3, even relatively simple compartmental and viral dynamics models may fail to satisfy global structural identifiability under commonly used observation schemes. These examples emphasize that identifiability is a property of the combined model–measurement system rather than of the differential equations alone.

A major contribution of `StructuralIdentifiability.jl` and the workflow presented in this tutorial is the integration of symbolic identifiability analysis directly into a modern scientific computing environment. The examples demonstrate how model formulation, simulation, and identifiability analysis can be performed within a unified and reproducible workflow. Several broader themes emerge from the study. First, measurement selection plays a critical role in determining parameter recoverability, and additional observations may restore identifiability. Second, identifiable parameter combinations often represent scientifically meaningful quantities rather than merely mathematical artifacts. Third, structural identifiability serves as a necessary prerequisite for

downstream statistical inference, including profile likelihood analysis, Bayesian calibration, and uncertainty quantification [1, 2, 28].

The present framework is restricted primarily to deterministic rational ODE models and does not directly address stochastic systems, delay equations, partial differential equations, or neural-network parameter identifiability. Furthermore, structural identifiability alone does not guarantee model adequacy or practical parameter recoverability under finite noisy data. Nevertheless, modern symbolic methods and open-source software have substantially reduced the historical barriers associated with identifiability analysis, making it increasingly feasible to incorporate these methods routinely into contemporary modelling workflows.

9. Conclusion

This tutorial presented a reproducible computational framework for symbolic structural identifiability analysis of mechanistic ordinary differential equation models using `StructuralIdentifiability.jl`. By combining mathematical theory, symbolic computation, and practical case studies within the Julia SciML ecosystem, the manuscript demonstrated how identifiability analysis can be integrated directly into modern modelling workflows. The examples illustrated globally identifiable systems, locally identifiable systems, structural non-identifiability, and identifiable parameter combinations, together with the role of additional measurements and reparameterization in restoring parameter recoverability.

More broadly, the tutorial highlights the importance of structural identifiability as a foundational prerequisite for reliable parameter estimation, mechanistic interpretation, and uncertainty quantification. Modern open-source tools now make symbolic identifiability analysis both accessible and computationally practical for researchers working in mathematical biology, epidemiology, pharmacokinetics, systems biology, and related disciplines. The central recommendation of this work is that structural identifiability analysis should become a routine step in the development and calibration of mechanistic ODE models, rather than a specialized analysis performed only after difficulties arise.

Acknowledgments

The author gratefully acknowledges the developers of `StructuralIdentifiability.jl` and the broader Julia SciML community, whose open source contributions made this work possible. The author also thanks colleagues at Delaware State University for productive discussions on identifiability in epidemic models.

Data and Code Availability

All Julia scripts, annotated notebooks, figures, and computational workflows used in this study are openly available under the MIT license. The complete reproducible repository associated with this manuscript has been permanently archived on Zenodo with a citable DOI [3]. The archived materials include the source code, example implementations, and supporting files required to reproduce all computational analyses and figures presented in this tutorial.

References

[1] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, and J. Timmer, "Structural and practical identifiability analysis of partially observed dynamical models by

exploiting the profile likelihood," *Bioinformatics*, vol. 25, no. 15, pp. 1923–1929, 2009. doi: https://doi.org/10.1093/bioinformatics/btp358.

[2] F.-G. Wieland, A. L. Hauber, M. Rosenblatt, C. Tönsing, and J. Timmer, "On structural and practical identifiability," *Current Opinion in Systems Biology*, vol. 25, pp. 60–69, 2021. doi: https://doi.org/10.1016/j.coisb.2021.03.005.

[3] A. Alsammani, "`aalsammani/Structural_Identifiability_Tutorial_Julia`: Structural Identifiability Tutorial with Julia v1.0.1," *Zenodo*, version 1.0.1, 2026. doi: https://doi.org/10.5281/zenodo.18684344.

[4] N. Tuncer and T. T. Le, "Structural and practical identifiability analysis of outbreak models," *Mathematical Biosciences*, vol. 299, pp. 1–18, 2018. doi: https://doi.org/10.1016/j.mbs.2018.02.004.

[5] R. Bellman and K. J. Åström, "On structural identifiability," *Mathematical Biosciences*, vol. 7, no. 3–4, pp. 329–339, 1970. doi: https://doi.org/10.1016/0025-5564(70)90132-X.

[6] H. Pohjanpalo, "System identifiability based on the power series expansion of the solution," *Mathematical Biosciences*, vol. 41, no. 1–2, pp. 21–33, 1978. doi: https://doi.org/10.1016/0025-5564(78)90063-9.

[7] C. Cobelli and J. J. DiStefano III, "Parameter and structural identifiability concepts and ambiguities: A critical review and analysis," *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology*, vol. 239, no. 1, pp. R7–R24, 1980. doi: https://doi.org/10.1152/ajpregu.1980.239.1.R7.

[8] E. Walter and L. Pronzato, *Identification of Parametric Models from Experimental Data*. Berlin, Germany: Springer, 1997. doi: https://doi.org/10.1007/978-3-642-57909-0.

[9] L. Ljung and T. Glad, "On global identifiability for arbitrary model parametrizations," *Automatica*, vol. 30, no. 2, pp. 265–276, 1994. doi: https://doi.org/10.1016/0005-1098(94)90029-9.

[10] H. Miao, X. Xia, A. S. Perelson, and H. Wu, "On identifiability of nonlinear ODE models and applications in viral dynamics," *SIAM Review*, vol. 53, no. 1, pp. 3–39, 2011. doi: https://doi.org/10.1137/090757009.

[11] G. Bellu, M. P. Saccomani, S. Audoly, and L. D'Angiò, "DAISY: A new software tool to test global identifiability of biological and physiological systems," *Computer Methods and Programs in Biomedicine*, vol. 88, no. 1, pp. 52–61, 2007. doi: https://doi.org/10.1016/j.cmpb.2007.07.002.

[12] A. F. Villaverde, A. Barreiro, and A. Papachristodoulou, "Structural identifiability of dynamic systems biology models," *PLoS Computational Biology*, vol. 12, no. 10, p. e1005153, 2016. doi: https://doi.org/10.1371/journal.pcbi.1005153.

[13] H. Hong, A. Ovchinnikov, G. Pogudin, and C. Yap, "Global identifiability of differential models," *Communications on Pure and Applied Mathematics*, vol. 73, no. 9, pp. 1831–1879, 2020. doi: https://doi.org/10.1002/cpa.21921.

[14] R. Dong, C. Goodbrake, H. A. Harrington, and G. Pogudin, "Differential elimination for dynamical models via projections with applications to structural identifiability," *SIAM Journal*

on Applied Algebra and Geometry, vol. 7, no. 1, pp. 194–235, 2023. doi: https://doi.org/10.1137/22M1469067.

[15] T. S. Ligon, F. Fröhlich, O. T. Chiş, J. R. Banga, E. Balsa-Canto, and J. Hasenauer, "GenSSI 2.0: Multi-experiment structural identifiability analysis of SBML models," *Bioinformatics*, vol. 34, no. 8, pp. 1421–1423, 2018. doi: https://doi.org/10.1093/bioinformatics/btx735.

[16] N. Meshkat, M. Eisenberg, and J. J. DiStefano III, "An algorithm for finding globally identifiable parameter combinations of nonlinear ODE models using Gröbner bases," *Mathematical Biosciences*, vol. 222, no. 2, pp. 61–72, 2009. doi: https://doi.org/10.1016/j.mbs.2009.08.001.

[17] H. Hong, A. Ovchinnikov, G. Pogudin, and C. Yap, "SIAN: Software for structural identifiability analysis of ODE models," *Bioinformatics*, vol. 35, no. 16, pp. 2873–2874, 2019. doi: https://doi.org/10.1093/bioinformatics/bty1069.

[18] O.-T. Chiş, J. R. Banga, and E. Balsa-Canto, "Structural identifiability of systems biology models: A critical comparison of methods," *PLoS ONE*, vol. 6, no. 11, p. e27755, 2011. doi: https://doi.org/10.1371/journal.pone.0027755.

[19] A. Ovchinnikov, A. Pillay, G. Pogudin, and T. Scanlon, "Computing all identifiable functions of parameters for ODE models," *Systems & Control Letters*, vol. 157, p. 105030, 2022. doi: https://doi.org/10.1016/j.sysconle.2021.105030.

[20] C. Rackauckas and Q. Nie, "DifferentialEquations.jl: A performant and feature-rich ecosystem for solving differential equations in Julia," *Journal of Open Research Software*, vol. 5, no. 1, p. 15, 2017. doi: https://doi.org/10.5334/jors.151.

[21] J. Bezanson, A. Edelman, S. Karpinski, and V. B. Shah, "Julia: A fresh approach to numerical computing," *SIAM Review*, vol. 59, no. 1, pp. 65–98, 2017. doi: https://doi.org/10.1137/141000671.

[22] A. Sedoglavic, "A probabilistic algorithm to test local algebraic observability in polynomial time," *Journal of Symbolic Computation*, vol. 33, no. 5, pp. 735–755, 2002. doi: https://doi.org/10.1006/jsco.2002.0516.

[23] Y. R. Liyanage, O. Saucedo, N. Tuncer, and G. Chowell, "A tutorial on structural identifiability of epidemic models using `StructuralIdentifiability.jl`," *arXiv preprint*, arXiv:2505.10517, 2025. doi: https://doi.org/10.48550/arXiv.2505.10517.

[24] W. O. Kermack and A. G. McKendrick, "A contribution to the mathematical theory of epidemics," *Proceedings of the Royal Society A*, vol. 115, no. 772, pp. 700–721, 1927. doi: https://doi.org/10.1098/rspa.1927.0118.

[25] A. S. Perelson, A. U. Neumann, M. Markowitz, J. M. Leonard, and D. D. Ho, "HIV-1 dynamics in vivo: Virion clearance rate, infected cell life span, and viral generation time," *Science*, vol. 271, no. 5255, pp. 1582–1586, 1996. doi: https://doi.org/10.1126/science.271.5255.1582.

[26] M. C. Eisenberg, S. L. Robertson, and J. H. Tien, "Identifiability and estimation of multiple transmission pathways in cholera and waterborne disease," *Journal of Theoretical Biology*, vol. 324, pp. 84–102, 2013. doi: https://doi.org/10.1016/j.jtbi.2013.01.032.

[27] A. F. Villaverde, A. Barreiro, and A. Papachristodoulou, "Observability and structural identifiability of nonlinear biological systems," *IEEE Control Systems Letters*, vol. 3, no. 4, pp. 1171–1176, 2019. doi: https://doi.org/10.1109/LCSYS.2019.2913725.

[28] K. E. Hines, T. R. Middendorf, and R. W. Aldrich, "Determination of parameter identifiability in nonlinear biophysical models: A Bayesian approach," *The Journal of General Physiology*, vol. 143, no. 3, pp. 401–416, 2014. doi: https://doi.org/10.1085/jgp.201311116.

[29] M. C. Eisenberg and M. A. L. Hayashi, "Determining identifiable parameter combinations using subset profiling," *Mathematical Biosciences*, vol. 256, pp. 116–126, 2014. doi: https://doi.org/10.1016/j.mbs.2014.08.008.

[30] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems*, vol. 31, 2018. doi: https://doi.org/10.48550/arXiv.1806.07366.

[31] G. K. Sandve, A. Nekrutenko, J. Taylor, and E. Hovig, "Ten simple rules for reproducible computational research," *PLoS Computational Biology*, vol. 9, no. 10, p. e1003285, 2013. doi: https://doi.org/10.1371/journal.pcbi.1003285.